

CFD Study of a Formula 1 Car Front Wing

From CAD Modelling to ANSYS Analysis, Mapping Downforce, Drag and Flap Stall Across Varying Flap Angles.

Justin Chee

University of Bristol

Bristol, UK/Singapore, Singapore

Student Email: yx25498@bristol.ac.uk

Personal Email: mrjustinchee@gmail.com

Abstract

The aerodynamic performance of high-speed vehicles like racing cars and fighter jets are strongly influenced by their aerodynamic components, that maximise downforce or lift while minimizing drag. In this study, the effects of varying flap angles in a multi-element F1 front wing on aerodynamic performance are investigated using integrated Computer-Aided-Design (CAD) modelling. Computational-Fluid-Dynamics (CFD) Simulations are then used to simulate and test how the downforce and drag are affected. The design comprises of a multi-element front wing designed in Fusion 360, which was then analysed using ANSYS Fluent using the $k-\omega$ Shear Stress Transport (SST) turbulence model under steady-state external flow conditions with the inlet velocity of 70m/s, approximately 252km/h, representative of an F1 car's straightline/cornering speed. The computational domain was dimensioned in accordance with standard aerodynamic domain-sizing practices, to minimise boundary effects and accurately capture downstream wake behaviours. An independent mesh study was then conducted to further support the findings to ensure that the results were reliable and independent of meshing. The results demonstrated a measurable relationship between flap angle and aerodynamic performance, showing the efficiency drop off as the angle increased beyond a certain point. This study demonstrates the effectiveness of a combined CAD-CFD workflow, providing a clear and accessible methodology to find forces acting on a multi element aerodynamic component during its preliminary design phase.

1. INTRODUCTION

A Formula 1 car is made up of many sub-components, like the front wing, tail fin, main chassis and wheels, with each part contributing different functions but all ultimately working together to ultimately improve the car's performance. In the wing of a Formula 1 car, aerodynamic forces are primarily determined by the pressure and velocity distributions along the surface of the wing, which in turn generates the forces along the wing. This report investigates the engineering processes involved to design the front wing

of a Formula 1 car, from CAD Modelling in Autodesk Fusion to CFD analysis in ANSYS. This study investigates how varying the flap angle of an F1 front wing affects the aerodynamic forces generated by the wing, with a strong focus on the downforce and drag produced.

2. METHODOLOGY

1. Geometry and Modelling Approach

The 3D model was constructed in Fusion 360, created with inspiration from the Mercedes 2025 W16 Front Wing, representative of modern Formula One cars. The model was created in a 1:1 ratio and measured to ensure adherence to the 2-meter width guidelines of the FIA 2025 car dimensions. Utilising publicly available top/front/side view images, drawings of the rough general shape were created and input into the Fusion where subsequent modelling was carried out. The modelling process began with the nose cone, followed by the main plane, end plates, turning vanes and flaps. For the flaps, three NACA-6412 airfoils were linked from the endplates to inner turning vanes using Fixed Point Splines. The flap angle was then parameterized through 6 different steps, from 0° degrees to 25° at 5° intervals. Each model was then exported as a STEP file, ready to be imported into ANSYS Spaceclaim. Below is an isometric view of the finished model of the wing, seen in Fig 1.

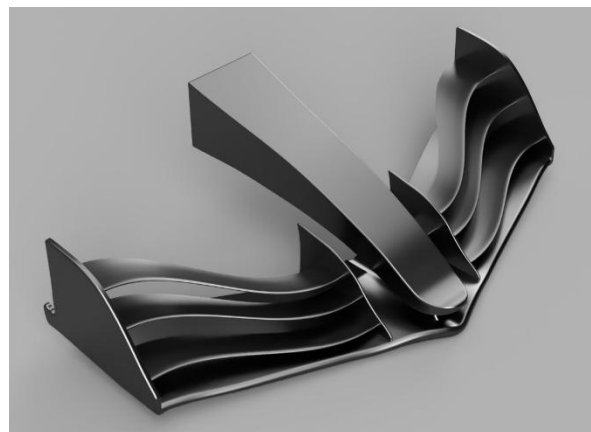


Figure 1: Isometric Fusion CAD Model of the Multi-element Wing.

II. Fluid Domain

Each STEP file was uploaded to Spaceclaim, where an enclosure was generated, according to standard external aerodynamics domain-sizing practices with reference to the length of the structure. This created an enclosure with the dimensions 3m upstream of the leading edge (5x structure length), 5.5m downstream (10x structure length), 3m to each side and 5.5m above the model. This ensured that the blockage domain remained <5% of the wing's frontal area and allowed for a sufficient space behind the model to allow wake and vortex development. The base of the enclosure was set at 75mm, simulating the actual clearance between the wing and the ground in an F1 car. The ground was later set in ANSYS Fluent as a moving wall, simulating the car's relative motion to a track instead of a stationary boundary.

III. Meshing

Within Spaceclaim, 3 more small bodies of influence were generated, one below the wing, one encompassing the first body and the wing, extended high above the wing, with the third body behind the wing, as seen in Fig 2. Due to the ANSYS Student license's 1,000,000-cell limit, some constraints had to be taken when generating the mesh, to ensure the element count did not exceed the limit. Global element sizing was set coarse at 300mm to minimize the far-field cost, with curvature and proximity based adaptive sizing enabled to ensure leading and trailing edges along with the narrow gaps between flaps were properly captured. Volumetric refinement was also added in the BOIs, initially targeted at 10mm, 30mm and 15mm per respective zones but due to the element limit set on the student license, this was increased to 20mm, 200mm and 100mm respectively. Once combined with the capture proximity and capture curvature however, this added an additional layer of refinement in the narrow gap and high curvature regions around the structure. This followed layers of troubleshooting and bug-fixing, generating an overall element number of roughly 900,000, just below the 1,000,000-cell limit. Following the mesh generation, the meshing workflow was updated on ANSYS Workbench to prepare the mesh for processing.

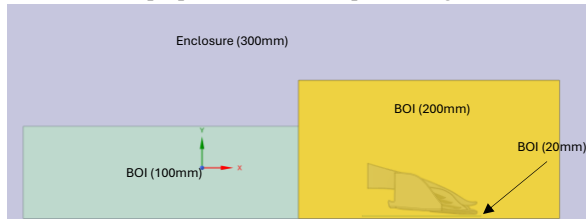


Figure 2: Bodies of Influence Locations for reference

IV. Setup

Within ANSYS Fluent, simulations were run using the k- ω SST turbulence model, chosen for its accurate prediction of flow separation from a smooth surface (ANSYS, 2026), which would be more prevalent in the multi-element wing flaps at higher incidence. This model is also widely used in Computational Fluid Dynamics to simulate turbulent flows characterized by chaotic motion and eddies. Air was chosen

as the fluid, modelled with default Fluent settings ($\rho=1.225\text{kg/m}^3$), with an inlet velocity of 70m/s, chosen to simulate straight-line and cornering speeds of 252km/h, representative of an F1 car's speeds in given situations. The pressure outlet was set at 0 gauge pressure, and all surfaces on the wing were defined as no-slip walls. Each case was initialized via hybrid initialization and run for 150 calculations to ensure monitored residuals and coefficients achieved converging trends, or until convergence itself was achieved. Streamlines were generated using Fluent's Pathlines function alongside the wing, to provide a visual of the airflow over the wing, seen in Fig 3. Lift and drag force values were then generated with reference to the wing, with the data shown in Fig 4.

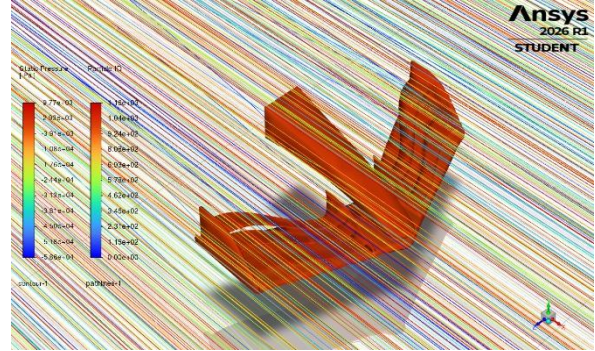


Figure 3: Pathlines Passing over Wing Model to Simulate Airflow

3. MESH INDEPENDENCE STUDY

A mesh independence study was conducted at the 0° baseline across three different cell densities, by varying the different Body of Influence element sizes, while keeping all other solver setting constant. The force values were recorded and presented as shown on Table 1.

Table 1: Force Magnitudes with Different Cell Counts.

Mesh	Cell Count	Downforce (N)	Drag Force (N)
Coarse	192233	1472.64	486.514
Baseline	749755	1538.71	464.231
Fine	921240	1553.97	446.037

From here, it can be observed that downforce increased by 4.49% between Coarse and Baseline, but sharply decreased towards the fine region, increasing only by 1% in the latter. Drag however, held a steady decrease with a 4.58% change between the coarse and baseline meshes and a 3.92% change between the baseline and fine meshes. This suggests that the drag prediction has not yet achieved the same level of mesh independence as downforce. This is possibly due to the face that drag is more sensitive to boundary separation, and a higher mesh may be required to observe these effects.

The change from baseline to fine in meshing was only 23%, compared to the coarse and baseline's difference of roughly 390%, which was caused primarily due to the ANSYS Student License's 1,000,000 cell limit. With the downforce (the primary variable of the study) having a level of mesh independence, due to the constraints in place by the cell limit, the baseline mesh was retained for the remaining 5 flap angles in the study.

4. RESULTS AND DISCUSSION

A. Results

The forces generated for each Flap Angle were then recorded onto a table as shown in Table 2.

Table 2: Forces Generated at Varying Flap Angles

Flap Angle	Down Force (N)	Drag Force (N)
0°	1538.7	464.2
5°	1936.4	602.3
10°	2354.1	825.9
15°	2845.8	1009.9
20°	3097.1	1165.1
25°	3341.5	1288.5

Using the data collected, the Coefficients of Lift and Drag can be calculated using Equation 1 and 2 (NASA, 2026).

$$Cl = \frac{L}{\frac{1}{2}\rho v^2 A} \quad [1]$$

$$Cd = \frac{D}{\frac{1}{2}\rho v^2 A} \quad [2]$$

Where L is the Lift force generated, D is the Drag Force generated, ρ is the density of air (1.225kgm/s) and A is the reference area of the wing. The reference area of the wing was calculated to be 0.930 m² by taking the top-down cross-sectional area of the wing. By using Equations 1 and 2, the Cl and Cd was calculated, with the results shown in Table 3.

Table 3: Forces Generated at Varying Flap Angles

Flap Angle	Lift Coefficient	Drag Coefficient
0°	0.55	0.166
5°	0.69	0.216
10°	0.84	0.296
15°	1.02	0.362
20°	1.11	0.417
25°	1.20	0.462

With these two sets of data, a graph could be plotted to demonstrate the relationship between the Coefficients and Flap Angles, to analyse for any trends. This is seen in Fig 4 and 5.

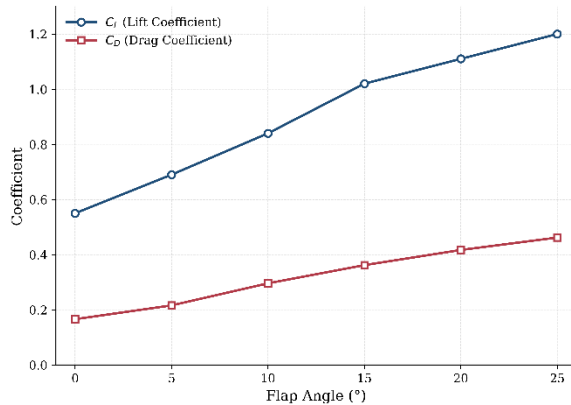


Figure 4: Graph of Flap Angles in relation to Coefficients

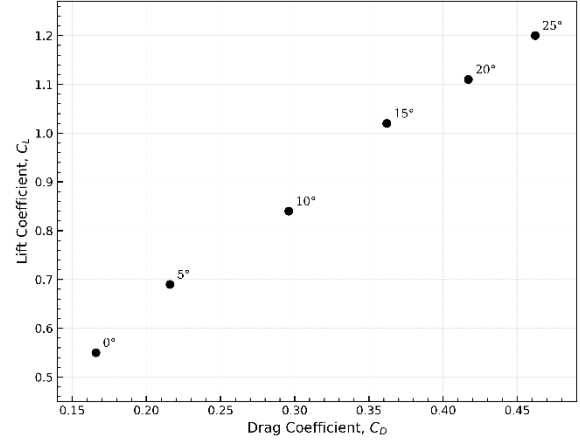


Figure 5: Cl vs Cd Polar Plot for Flap Angles 0° to 25°.

From this, it is observable that Cl and Cd both increased across all flap angles, with the Coefficient of Lift more than doubling from 0° to 25°. However, this rate of increase is not constant in the Lift Coefficient, with the slope more than halving after 15°.

B. Discussion

The lift-curve slope drop-off from 15° to 25° showed a pattern of Cl growth at a substantially reduced rate, consistent with the onset of progressive flow separation on flap elements, as opposed to an abrupt stall. This is because in a hard stall, the Cl tends to drop sharply after exceeding the critical angle, rather than a sustained (but slower) Cl growth. This suggests that separation occurred gradually across the flow surface, rather than causing a wide breakdown in attached flow. The Cl/Cd Polar Plot in Fig 5 is consistent with this theory, as rather than showing a sharp drop associated with a hard stall, the polar plots trace a smooth, non-linear path across the full angle range. The absence of a sharp drop off suggests a gradual efficiency loss over a stall with total separation and is a milder signal than the lift curve alone can demonstrate. The L/D also declined monotonically from 3.31 at 0° to 2.59 at 25°, with rate of decline increasing after 15°, once again supporting the gradual loss in efficiency over a sudden complete stall.

This study did not include direct flow field visualization, and as such the separation mechanism itself was not directly observed. The force-based evidence points towards a form of gradual flow separation between 15° and 20°, though the smoothness of the polar plot suggests that it may be a gradual efficiency loss instead of a definitive hard-stall. This distinction however can only be confirmed through direct flow visualization.

C. Limitations

Some simplifications were made in this experiment that should be considered when interpreting the results. Firstly, the height of the main plate was set at 75mm above ground throughout the experiment, which is not always true as it varies with aerodynamic load, fuel burn and many other factors. Secondly, the Bodies of Influence regions were

heavily coarsened by about 2-7 times due to ANSYS Student License's 1,000,000-cell limit in meshing. This would have limited the resolution of the underwing gap especially, affecting the venturi-driven downforce calculations this study was aimed to characterize. Finally, the freestream velocity of 70m/s was treated as fully incompressible, when at such speeds some compressibility effects may have been present and were not captured.

5. WHAT I'D DO NEXT

There are several ways that this study can be further developed. To further analyse the airflow and separation, a deeper dive into the airflow behaviour through transient simulations can be done, capturing vortex shedding or oscillating separations that are otherwise missed. A second study can be done on ride height behaviours, as by varying the ride height between 50-100mm, a relationship can be found between downforce and clearance, allowing for a deeper development of requirements and planning around what matters for a real car. Lastly, a full car integration will also be very beneficial as at least a simple car body will demonstrate turbulence behaviour in the presence of tire wakes caused by the open wheels and how the front wing can influence underbody flow as well.

6. BIBLIOGRAPHY

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